Adaptive Advance-Time Control for nFAPI Split under Unpredictable Latency

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Abstract—
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I. Introduction

A. Background and Motivation

With the evolution of 5G and the Open Radio Access Network (O-RAN), architectures based on functional splits have become the key to improving network flexibility and reducing deployment costs. Currently, the most mainstream implementation promoted by the O-RAN Alliance is Split Option 7.2 (Split 7.2). Because Split 7.2 divides the Physical (PHY) layer into high and low parts and transmits continuous in-phase and quadrature (IQ) samples, it imposes extremely high requirements on the fronthaul network.

Compared to Split 7.2, the 3rd Generation Partnership Project (3GPP) Split Option 6 (Split 6), which separates the network at the Media Access Control (MAC) and PHY layers, offers a more flexible alternative. First, Split 7.2 relies on hardware-based synchronization protocols, such as IEEE 1588 Precision Time Protocol (PTP) and Synchronous Ethernet (SyncE), to achieve nanosecond-level phase and time synchronization. Its timing window requirement is typically strict, often needing to be less than 100 μs. In contrast, the nFAPI (Network Functional Application Platform Interface) protocol, which is a prominent implementation of Split 6, employs a message-based synchronization mechanism. Through its proprietary P7 Node Sync procedure, nFAPI achieves slot-level synchronization (millisecond-level) and provides a highly flexible timing window ranging from 0 to 30,000 μs. Furthermore, in terms of payload, Split 6 transports packetized MAC Protocol Data Units (PDUs) rather than continuous IQ samples. These characteristics allow Split 6 to demonstrate better adaptability in non-ideal fronthaul network environments where high jitter from operating system (OS) scheduling and network delivery is present.

This paper chooses Split 6 as the primary research subject due to its significant deployment benefits. As indicated by Pérez-Romero et al. [1], Split 6 effectively reduces the traffic burden on the fronthaul network. Compared to low-layer split schemes that transmit frequency-domain signals, its bandwidth requirement is significantly lower. In addition, Khan et al. [2] demonstrate that for the deployment of sub-6 GHz small cell networks, Split 6 can maintain robust transmission performance over non-ideal fronthaul connections, thereby achieving an optimal cost-revenue trade-off.

Deploying the VNF and PNF on different machines introduces significant challenges. Because the MAC scheduler is separated from the PHY layer, strict slot synchronization is required between these distributed components. However, different traffic loads and non-ideal network deployments introduce unpredictable, non-deterministic latency. Consequently, the VNF cannot reliably transmit data exactly at the slot boundary, and existing open-source solutions suffer from synchronization failures when encountering unpredictable latency from routers and switches. This paper defines these latency and synchronization problems under non-ideal fronthaul networks and proposes a dynamic trigger scheduler adjustment algorithm. A detailed analysis of this non-deterministic latency and the corresponding verification of these problems will be presented in Section IV, specifically isolating these issues in Scenario I (IV-B).

B. Related Works

C. Contributions

The main contributions of this paper are summarized as follows.

- Contribution 1.
- Contribution 2.
- Contribution 3.

The rest of the paper is organized as follows. Section II defines the system model/architecture considered in this paper. The proposed analytical model/method is given in Section III. Section IV shows the numerical results. Concluding remarks are given in Section VI.

II. System Model/Architecture

Before diving into the system details, this study establishes a fundamental assumption: currently, there is a lack of Open Radio Access Network (O-RAN) Radio Units (O-RUs) in the market that natively support the Network Functional Application Platform Interface (nFAPI) via Split 6. As a result, the proposed system adopts a resilient two-stage split architecture. The network first connects via the Split 6 nFAPI interface between the network functions, and then translates to the standard O-RAN Split 7.2x interface to connect to the remote O-RU. Figure 1 illustrates this proposed system architecture and its delay management mechanism.

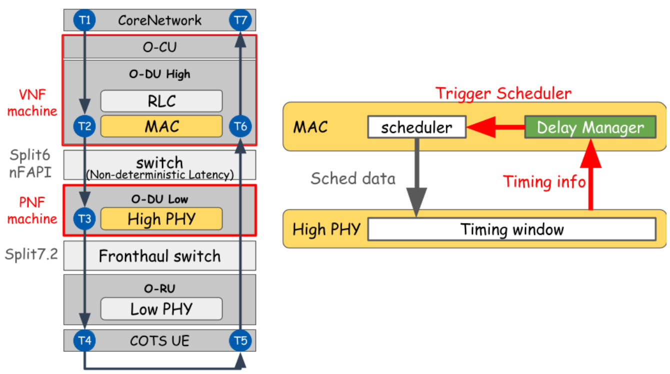

Fig. 1. System architecture and delay management mechanism.

A. A. End-to-End Architecture

The system establishes a complete 5G end-to-end communication link, extending from the core to the user equipment:

- Core Network (CN): The central element responsible for routing, session management, and authenticating user data.
- O-RAN Central Unit (O-CU): A logical node that handles higher-layer protocol stacks, such as the Radio Resource Control (RRC) and Packet Data Convergence Protocol (PDCP). It connects with distributed units via the F1 interface.
- O-RAN Distributed Unit (O-DU): To optimize computational performance and deployment flexibility, the system separates the O-DU into an O-DU High and an O-DU Low. The O-DU High is executed as a Virtualized Network Function (VNF) on a virtualized machine, whereas the O-DU Low is deployed as a Physical Network Function (PNF) on dedicated hardware.
- Switch: A commercial network switch is located between the VNF and PNF. This mid-layer connection utilizes the nFAPI protocol. However, traversing this packet-switched network inevitably generates non-deterministic latency.
- O-RAN Radio Unit (O-RU): A specialized hardware unit responsible for low-level physical layer (Low PHY) processing, digital beamforming, and Radio Frequency (RF) transmission.
- Commercial Off-The-Shelf User Equipment (COTS UE): Standard commercial smartphones or modems that act as the terminal receivers in the network.

B. B. O-DU High: MAC and Delay Manager

Within the Media Access Control (MAC) layer of the O-DU High, the core components handling timing mechanisms include the Scheduler and the nFAPI Delay Manager:

- Scheduler: Critical for allocating the available radio resources (such as time-frequency resource blocks) and determining the exact priority and timing of downlink/uplink data transmissions.
- nFAPI Delay Manager: This resides as the crucial control unit of this delay-aware architecture. Because unpredictable, unstable latency exists in the switch between the VNF and PNF, the Delay Manager continuously receives timing feedback from the lower physical layer. It uses this to dynamically calculate network jitter and proactively trigger the scheduler. This calculation guarantees that the scheduling instructions (`Sched data`) are generated ahead of time to account for network delays.

C. C. O-DU Low: High PHY and Timing Window

The O-DU Low manages the higher-level baseband processing of the physical layer (High PHY). At this level, the most critical synchronization mechanism is the nFAPI PNF Timing Window:

- Timing Window: A predefined, strict time duration. When the scheduling message (specifically the `DL_TTI.request`) sent by the O-DU High propagates across the network, it must accurately arrive and fall exactly within this Window. Otherwise, the High PHY will reject the message and fail to process the subframe data.
- Timing Info: The PNF continuously monitors the exact arrival timestamp of these control messages and periodically sends this "Timing Info" back to the Delay Manager in the MAC layer. This establishes a closed-loop feedback control system, compensating for the unpredictable transmission latency variations between the VNF and PNF machines.

D. D. Key Validation Metrics

To evaluate the stability, synchronization accuracy, and real-time performance of the proposed algorithm under non-ideal environments, this study defines the following three Key Validation Metrics, where RTT denotes the Round-Trip Time:

- Ping RTT ($T_7 - T_1$): Measures the end-to-end round-trip latency at the application layer. This metric captures the complete duration from when a packet is sent from the CN to when its return acknowledgment is received by the sender.
- HARQ RTT ($T_6 - T_2$): Measures the Hybrid Automatic Repeat Request (HARQ) loop delay specifically at the MAC layer. HARQ RTT accurately reflects the feedback efficiency and operational health of the link layer.
- VNF-PNF Downlink Latency ($T_3 - T_2$): Measures the one-way downlink latency accumulated from the virtualized machine to the physical machine. This metric is used to directly quantify and analyze the delay jitter impact injected by the intermediate Ethernet switch.

The following assumptions were made in this paper.

- Assumption 1.
- Assumption 2.
- Assumption 3.

III. Proposed Method

IV. Experimental Results

To validate the hypothesis and the effectiveness of the proposed dynamic trigger scheduler algorithm, we systematically designed multiple experimental scenarios. Section IV-A outlines the general software and hardware environment used across all tests. Subsequent sections present specific scenarios: Scenario I (IV-B) isolates and verifies the problems caused by non-deterministic latency under standard OpenAirInterface implementations, while Scenario II (IV-C) and subsequent experiments evaluate the robustness of our proposed time-advance control under varying levels of congestion.

A. A. Experimental Scenario

The testbed environment connects a series of high-performance servers, radio units, and a commercial smartphone. The detailed software and hardware configurations for the end-to-end network deployment are outlined below.

Software Configuration:

- Core Network (CN): Open5GS.
- nFAPI VNF (O-DU High): OpenAirInterface (branch: `2026.w13`), configured with 30kHz subcarrier spacing (SCS), Time Division Duplex (TDD) pattern 3D1S1U, and 4-layer Multiple-Input Multiple-Output (MIMO).
- nFAPI PNF (O-DU Low): OpenAirInterface (branch: `2026.w13`), utilizing a 4T4R (4 Transmitters, 4 Receivers) configuration.
- Fronthaul Version: O-RAN F release specifications.
- O-RU: Pegatron RU.
- COTS UE: Samsung Galaxy S20.

Hardware Configuration:

- VNF Server (O-DU High Host): Intel® Xeon® Gold 6226R Processor.
- PNF Server (O-DU Low Host): Intel® Xeon® Gold 6433N Processor.

B. B. Scenario I: Analysis of Non-Deterministic Latency

Deploying the VNF and PNF on different machines introduces significant challenges. Because the MAC scheduler is separated from the PHY layer, strict slot synchronization is required between these distributed components. However, even when deployed in the exact same network environment, different traffic loads directly impact the latency. Figure 2 illustrates the latency between the VNF scheduler and the PNF under varying traffic loads. As traffic increases, three factors heighten the overall delay: the scheduler requires more processing time, the data packing time lengthens, and the VNF-to-PNF One-Way Delay (OWD) is affected. Consequently, the VNF cannot reliably transmit data exactly at the slot boundary. Instead, it must dynamically advance its scheduling point to compensate for these varying delays and ensure packets arrive at the PNF on time.

To lower deployment costs, operators often want to deploy Split-6 nFAPI gNBs over non-ideal networks. However, beyond traffic load variations, these non-ideal networks introduce additional, unpredictable latency. Figure 3 shows the latency jitter caused by intermediate routers and switches in a non-ideal deployment. Existing open-source solutions, such as OpenAirInterface (OAI), provide basic nFAPI support but were

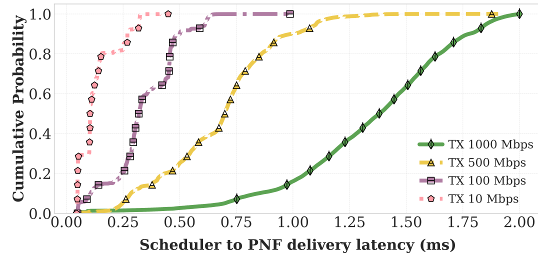

Fig. 2. Observation of latency between VNF scheduler and PNF under different traffic loads.

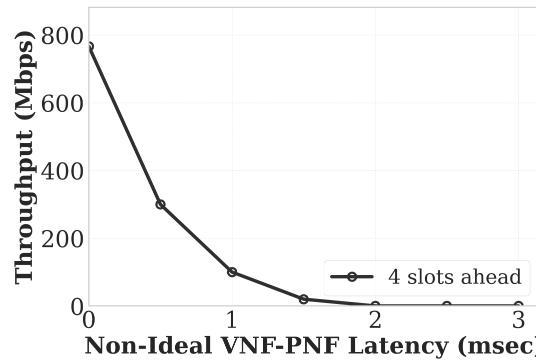

Fig. 3. Impact of unpredictable network latency in a non-ideal environment.

not designed to fully comply with the nFAPI specification's delay management procedures [3]. Therefore, they suffer from synchronization failures when encountering unpredictable latency from routers and switches. To address these challenges, this paper proposes a dynamic trigger scheduler adjustment algorithm that fully complies with the specification. Through systematic experiments, we verify the effectiveness of this dynamic adjustment and its benefits to overall network reliability.

C. C. Scenario II: [Placeholder for Future Experiment]

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